Cover Letter

Outlier,
360 22nd St, Suite 600, Oakland, CA 94612, United States

Dear Hiring Manager,

As a PhD in Theoretical Mathematics from Université Paris-Ést, I am eager to apply my expertise in combinatorial systems and algorithm design to refine generative AI models at Outlier. My academic research (peer-reviewed, arXiv :2110.12629), technical writing (LaTeX/arXiv), and teaching experience align perfectly with your need for experts to evaluate AI-generated content for accuracy and domain-specific coherence.

Key strengths include :

- **Analytical Rigor :** Proven through publications and a thesis on cylindric plane partitions.
- **Clear Communication :** Demonstrated via tutorials, academic papers, and the TindArt app. My first language is English.
- **Remote Collaboration :** Freelance app development ensures flexibility and self-directed productivity.

I welcome the opportunity to enhance AI systems through factuality assessments and response ranking. Thank you for considering my application.

Sincerely, Robin Langer
phone : 0413 710 989 (Australia),
email : langer.robin@gmail.com

Robin Langer

Email : langer.robin@gmail.com
Phone : 0413 710 989
Address : Australia



Languages

— **Native Speaker :** English.
— **Other languages :** French, Thai.

Education

Doctorat (PhD)
Université Paris-Est, Ecole Doctorale MSTIC.
2011 – 2013
Thesis : ”Cylindric Plane partitions, alternating sign matrices and semi- circular systems”.
Available on arXiv : arXiv:2110.12629
Thesis advisor : Prof. Philippe Biane.

Early Stage Researcher (Theoretical Physics)
Université Paris 6, LPTHE.
May 2009 – August 2009
Marie Curie Research Training Network
ENRAGE (MRTN-CT-2004-5616)
Advisor : Prof. Paul Zinn-Justin

Masters Degree (Mathematics
The University of Melbourne
Grade : First Class Honours 2006 - 2008
Thesis : ”Macdonald Polynomials and Symmetric functions”.
Available on arXiv : arXiv:0907.3950
Thesis advisor : Prof. Ole Warnaar

Honours Year (Mathematics
The University of Melbourne
Grade : First Class Honours 2004 - 2005
Thesis : ”Symmetric Graphs and their Quotients”.
Available on arXiv : arXiv:1306.4798
Thesis advisor : Prof. Sanming Zhou

Experience

App Development (Flutter)
Enspyr (TindArt) 2025–Present

— The user is given a random sequence of artwork image and can swipe right or left depending on if they like the artwork or not. In this way they build up a private “digital art” collection
— The app will then aggregate data from multiple users and make use of a “recommender algorithm” to suggest new artwork that the user may enjoy.
— This app is available on the app store for both Android and iPhone.

Teaching (Tutorials)
Université Paris-Est, Ecole Doctorale MSTIC. 2011 – 2013

— Guiding students through problem solving strategies
— Correcting homework exercises
— Pedagogy

Skills

— Analytical thinking & Problem solving
— Breaking down abstract concepts into digestible components
— Technical Writing & Proofreading (LaTeX)
— Attention to detail & Patience and Persistance

References

Dr. Matthieu Josuat-Vergès
Chargé de recherche CNRS
IRIF – Université Paris-Cité
matthieu.josuat-verges@irif.fr
Tél. : +33 1 57 27 92 59

Dr. Ekaterina Vassilieva
Chargée de recherche CNRS
LIX – École Polytechnique
katya@lix.polytechnique.fr
Tél. : +33 1 69 33 41 24